

(5)

$$5. \quad x + y = k.$$

$$\min = x^2 + y^2$$

$$\min = x^2 + (k-x)^2$$

$$\min = x^2 + k^2 - 2kx + x^2$$

$$\min = 2x^2 - 2kx + k^2$$

$$(\min)' = 4x - 2k$$

$$\text{set } (\min)' = 0$$

$$4x - 2k = 0$$

$$x = \frac{k}{2} \Rightarrow y = \frac{k}{2}$$

$$\therefore \min = x^2 + y^2$$

$$\min = \left(\frac{k}{2}\right)^2 + \left(\frac{k}{2}\right)^2$$

$$= \frac{2k^2}{4}$$

$$\min = \frac{k^2}{2}$$

$\therefore \min$ is at least $\frac{1}{2}k^2$.